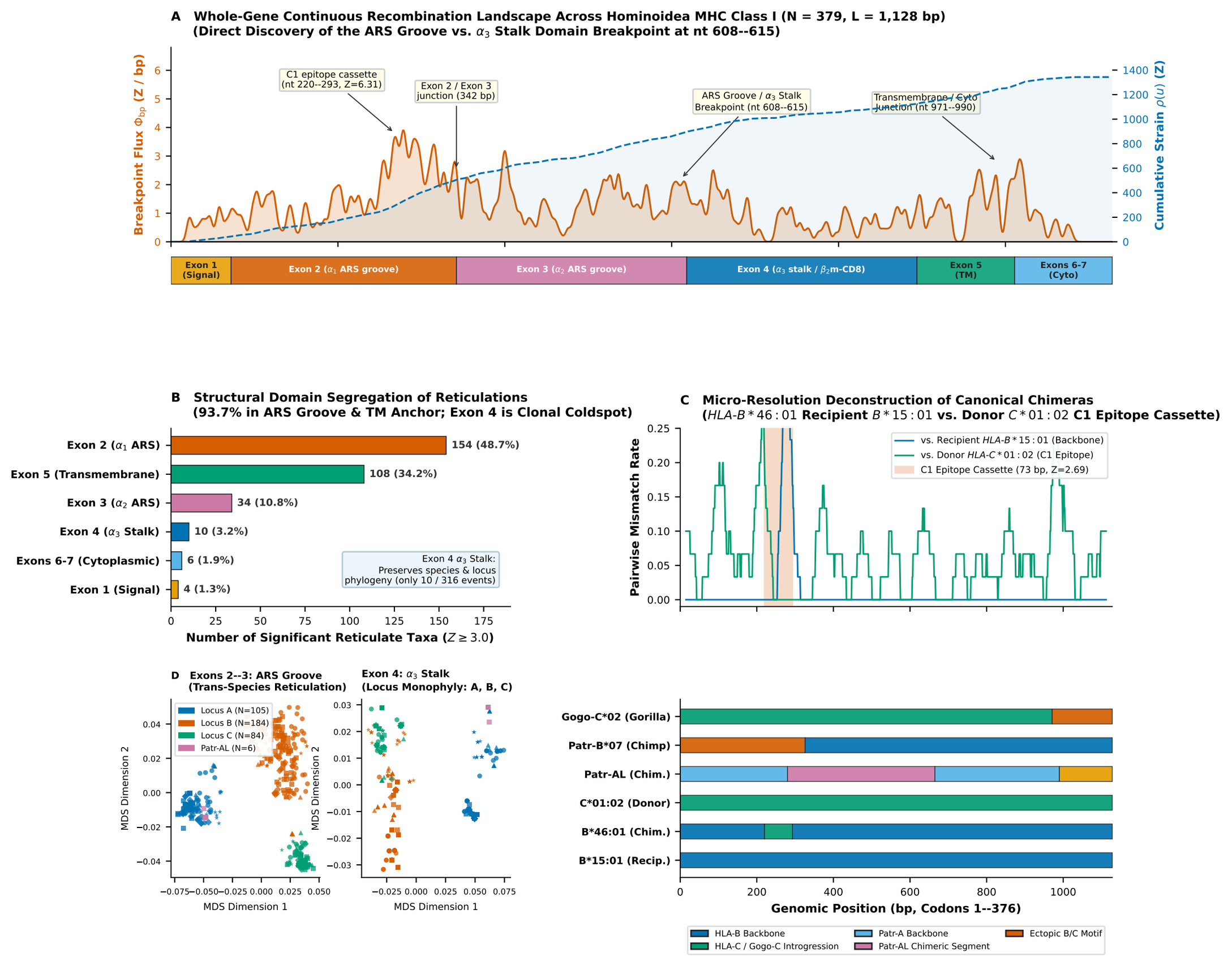

A Whole-Gene Continuous Recombination Landscape Across Hominoidea MHC Class I (N = 379, L = 1,128 bp)
(Direct Discovery of the ARS Groove vs. α3 Stalk Domain Breakpoint at nt 608--615)
Breakpoint Flux Φbp (Z / bp)
Cumulative Strain ρ(u) (Z)
C1 epitope cassette (nt 220--293, Z=6.31)
Exon 2 / Exon 3 junction (342 bp)
ARS Groove / α3 Stalk Breakpoint (nt 608--615)
Transmembrane / Cyto Junction (nt 971--990)
Exon 1 (Signal)
Exon 2 (α1 ARS groove)
Exon 3 (α2 ARS groove)
Exon 4 (α3 stalk / β2m-CD8)
Exon 5 (TM)
Exons 6-7 (Cyto)
B Structural Domain Segregation of Reticulations
(93.7% in ARS Groove & TM Anchor; Exon 4 is Clonal Coldspot)
Exon 2 (α1 ARS)
Exon 5 (Transmembrane)
Exon 3 (α2 ARS)
Exon 4 (α3 Stalk)
Exons 6-7 (Cytoplasmic)
Exon 1 (Signal)
154 (48.7%)
108 (34.2%)
34 (10.8%)
10 (3.2%)
6 (1.9%)
4 (1.3%)
Exon 4 α3 Stalk: Preserves species & locus phylogeny (only 10 / 316 events)
Number of Significant Reticulate Taxa (Z ≥ 3.0)
C Micro-Resolution Deconstruction of Canonical Chimeras
(HLA-B*46:01 Recipient B*15:01 vs. Donor C*01:02 C1 Epitope Cassette)
Pairwise Mismatch Rate
vs. Recipient HLA-B*15:01 (Backbone)
vs. Donor HLA-C*01:02 (C1 Epitope)
C1 Epitope Cassette (73 bp, Z=2.69)
D Exons 2--3: ARS Groove
(Trans-Species Reticulation)
Exon 4: α3 Stalk
(Locus Monophyly: A, B, C)
Locus A (N=105)
Locus B (N=184)
Locus C (N=84)
Patr-AL (N=6)
MDS Dimension 1
MDS Dimension 2
Gogo-C*02 (Gorilla)
Patr-B*07 (Chimp)
Patr-AL (Chim.)
C*01:02 (Donor)
B*46:01 (Chim.)
B*15:01 (Recip.)
Genomic Position (bp, Codons 1--376)
HLA-B Backbone
HLA-C / Gogo-C Introgression
Patr-A Backbone
Patr-AL Chimeric Segment
Ectopic B/C Motif